

Areal Organization of the Posterior Parietal Cortex of the Ferret (*Mustela putorius*)

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On grounds of electrophysiological mapping, cytoarchitecture, myeloarchitecture and callosal and thalamic connectivity, we have identified two cortical areas in the posterior parietal cortex of the ferret: posterior parietal caudal and rostral (PPc and PPr). These areas occupy the lateral and suprasylvian gyri, from the cingulate sulcus (medially) to the suprasylvian sulcus (laterally) and lie between visual areas 18 and 21 (posteriorly) and the somatosensory areas (anteriorly). Within both areas a coarse representation of the visual field was found and within PPr there was also a representation of the body. Each representation mirrors those within neighboring areas. Cytoarchitectonic and myeloarchitectonic fields within this cortical region did not correspond in any simple way to the physiological representations. The architectonic differences correlate to differential callosal connectivity, with predominant connectivity corresponding to the upper hemifield/head representations. PPr and PPc receive thalamic projections from a different, but overlapping, complement of thalamic nuclei. The superimposition of somatic and visual maps in PPr might relate to the probable role of this area in transforming retinal-centered to body-centered spatial coordinates. The organization of the parietal areas in the ferret resembles that of the flying fox and might unveil a common organizational plan from which the primate posterior parietal cortex evolved.

Introduction

The posterior parietal cortex, in concert with premotor cortex, plays an important role in a variety of complex visuomotor behaviors (Wise *et al.*, 1997). Together, these regions play an important role in spatial working memory tasks (Chafee and Goldman-Rakic, 1998) and in attentional mechanisms as seen in modulated responses of single units (Mountcastle, 1995) and as indicated by the hemi-neglect syndrome that follows lesion (Kerkhoff, 2001).

It is clear that parietal cortex has undergone a major expansion in primates compared with other mammals (Brodmann, 1909; Johnson 1990; Rosa *et al.*, 1999), with the emergence of at least 10 areas (Seltzer and Pandya, 1980; Pandya and Seltzer, 1982; Baizer *et al.*, 1991; Geyer *et al.*, 2000; Lewis and Van Essen, 2000a,b); however, data in other species are limited. In the flying fox, two physiologically distinct cortical areas have been defined (Rosa, 1999) and are termed the occipitoparietal area (OP) and posterior parietal area (PP), in the cytoarchitectonic field defined as 5/7 by Brodmann (Brodmann, 1909). Subdivision of this region in carnivores has been limited mainly to cytoarchitectonic studies in the kinkajou (Brodmann, 1909) and cat (Gurewitsch and Chatschaturian, 1928; Hassler and Muhs-Clement, 1964; Sanides and Hoffman, 1969; Heath and Jones, 1971); all of these studies present different interpretations; however, connectional studies (Marcotte and Updyke, 1982; Symonds and Rosenquist, 1984; Olson and Lawler, 1987) and a recent physiological study (Pigarev and Rodionova, 1998) have confirmed the visual nature of this region.

Despite these studies, it is unclear if a common set of organizational principles exist for this region and, if there are such principles, what trends might have been expressed in the evolution of the complex of parietal areas in the primate. These uncertainties and the lack of an adequate parcellation of this region in the carnivores motivated the present study, which is part of a broader project aimed at defining the cortical visual system of the ferret as a model for investigating issues of developmental plasticity at the system level (Innocenti *et al.*, 2002).

Preliminary findings in studies of the occipital visual areas of the ferret (Innocenti *et al.*, 2002; Manger *et al.*, 2002) led us to postulate that in the region between extrastriate areas 18 and 21 and the somatosensory cortex of the ferret, several visual or visual/somatosensory parietal areas might exist. To examine these possibilities we used physiological mapping, cytoarchitecture, myeloarchitecture and thalamocortical and callosal connectivity. Here we provide the first detailed description of posterior parietal cortex in a species other than the macaque monkey.

Materials and Methods

Twelve adult female ferrets (*Mustela putorius*) weighing between 600 and 1000 g were used. All experiments were performed according to Swedish and European Community guidelines for the care and use of animals in scientific experiments.

Three series of experiments were carried out, for electrophysiological mapping ($n = 4$), architectural analysis ($n = 2$) and anatomical connections ($n = 6$). Our methods for architectonics, electrophysiological mapping and anatomical tracing have been described previously (Innocenti *et al.*, 2002; Manger *et al.*, 2002), therefore here we detail mostly what was specific to the present series of experiments.

Electrophysiological Mapping

As previously described (Manger *et al.*, 2002), the animals were initially anesthetized with i.m. doses of ketamine hydrochloride (Ketalar, 10 mg/kg) and medetomidin hydrochloride (Domitor, 0.08 mg/kg), supplemented with atropine sulfate (0.15 mg/kg). An i.v. cannula was secured into the femoral vein and a tracheal tube (5 mm diameter) inserted. The animal was placed in a stereotaxic frame, artificially ventilated and anesthetized via the tracheal tube with 1% isoflurane in a mixture of 1:1 nitrous oxide and oxygen, with expired CO₂ maintained between 3.5 and 4%. Body temperature was maintained between 37 and 38°C, and heart rate monitored to ensure stability of the animal. The animal was paralyzed with an initial i.v. bolus of pancuronium bromide (0.15 mg/kg) and paralysis was maintained with a continuous infusion of pancuronium bromide (6 µg/kg/h).

Multi-unit activity was recorded with varnish-isolated tungsten microelectrodes (impedance between 0.95 and 1.3 MΩ) inserted perpendicularly in the exposed parietal cortex. The electrophysiological signal was amplified and filtered, visualized on an oscilloscope and played through a loudspeaker. Recordings were made at distances of between 500 and 1000 µm below the pial surface. Each penetration in a given antero-posterior row of recordings was separated by between 250 and 500 µm, dependent on the cortical vasculature. Each recording row was separated medio-laterally by 700–1400 µm. Visual receptive fields were

mapped with stationary or moving luminous white circles varying between 1 and 20° diameter and were plotted on a hemispheric screen in front of the animal, on which the optic disc had been ophthalmoscopically projected. Somatic responses were evoked with light touch, tapping or manipulation of the articulations and the receptive fields plotted on standard figurines of the animal body surface.

At the conclusion of the recording session, the animal was killed and perfused and the brain treated and cryoprotected as described below. The hemisphere from which we recorded was partially manually flattened (Manger *et al.*, 2002) and 50 µm sections were cut on a freezing microtome in a plane parallel to the pial surface. Alternate sections were treated for CO reactivity or myelin. For the reconstruction of recording sites, the sections were examined with a stereoscope and the outlines of the section, larger blood vessels, tissue artifacts and recording sites marked on a camera-lucida drawing. Architectonic boundaries, particularly that of the anterior border of area 18 (Manger *et al.*, 2002) and the posterior border of 3b, were determined from the CO staining at this stage.

For the determination of the physiological boundaries, several criteria were employed simultaneously. These included reversals in receptive field topographical progressions, changes in the size of the receptive field and changes in the neuronal response properties. Manual reconstruction of the maps, as is standard in studies of this kind (Rosa, 1999), was undertaken.

Architectural Analysis

Architectural studies were performed in four hemispheres. The animals were given a lethal dose of pentobarbital (80 mg/kg, i.p.) and perfused intracardially, initially with a rinse of 0.9% saline solution (4°C, 500 ml/kg), followed by fixation with 4% paraformaldehyde in 0.1 M phosphate buffer (4°C, 1000 ml/kg). The brain was removed and post-fixed overnight in the same solution, then transferred into a 30% sucrose solution in 0.1 M phosphate buffer (at 4°C) and allowed to equilibrate. The hemispheres were sectioned (50 µm thick sections) in two planes, coronal ($n = 2$) and normal to the dorsolateral surface of the cortex ($n = 2$). Serial sections were stained with cresyl violet or myelin (Gallyas, 1979), observed and photographed by low- and high-power microscopy.

Injections of Tract Tracers

The animals were initially anesthetized as described above and placed in a stereotaxic frame. A mixture of 1% isoflurane in 1:1 nitrous oxide and oxygen was delivered through a mask, while the animal maintained its own respiration. Anesthetic level was monitored using the eye blink and withdrawal reflexes, in combination with measurement of the heart rate. The parieto-occipital cortex was exposed under aseptic conditions. A few electrophysiological recordings were made to ensure placement of the tracer at specified retinotopic locations within the two cortical areas defined in the mapping experiments. Approximately 500 nl of tracer was delivered at each injection site through a Hamilton microsyringe. For the investigation of callosal connections, we used 10% wheatgerm agglutinin-horseradish peroxidase (WGA-HRP; Sigma) dissolved in phosphate buffer (0.1 M, pH 7.35). Several injections were made across the extent of posterior parietal cortex. For the investigation of thalamocortical connections, we used two fluorescent dextran tracers, fluororuby and fluoroemerald (5% in 0.1 M phosphate buffer; Molecular Probes). In each animal, one injection of each tracer was applied in two combinations. First, in the same cortical area, with one injection in the lateral gyrus and one in the suprasylvian gyrus ($n = 2$). Secondly, two injections were placed in the suprasylvian, or lateral gyrus, but in different cortical areas ($n = 2$). Thus, a total of eight injections was delivered, four in each cortical area.

After completion of injections, a soft contact lens was cut to fit over the exposed cortex, the retracted dura pulled over the contact lens, and the excised portion of bone repositioned and held in place with dental acrylic. The temporal muscle was reattached using surgical glue and the midline incision of the skin sutured. Antibiotics were administered in all cases (Terramycin, 40 mg/kg, each day for 2–5 days). These animals were given a 3 day (WGA-HRP cases), or 2 week (dextran cases) recovery period to allow for tracer transport.

At the end of the recovery period, the animals were given a lethal dose of pentobarbital (80 mg/kg, i.p.) and, in the dextran cases, perfused as

described above. For the WGA-HRP cases, 3% paraformaldehyde was used as the fixative. The cerebral hemispheres were dissected from the remainder of the brain and cut in a semi-flattened plane, parallel to the pial surface as previously described (Manger *et al.*, 2002). Alternate 50 µm thick sections were stained for CO, or mounted on gel-coated slides for fluorescent microscopy. In one callosal case the brain was sectioned in a coronal plane. The thalamus and midbrain were sectioned (50 µm) in the coronal plane and serial sections stained for CO or Nissl, or mounted for fluorescent microscopy. The labeled cells were plotted using either the Neurolucida system (Innocenti *et al.*, 2002), or a camera lucida. Architectonic details of the cortical areas and thalamic nuclei were determined and drawn onto the plots of the labeled cells.

Results

On grounds of electrophysiological mapping and connectivity, we delineated two cortical areas in the posterior parietal cortex of the ferret, posterior parietal caudal (PPc) and posterior parietal rostral (PPr). Both areas occupy the crowns of the lateral and suprasylvian gyri and both banks of the lateral sulcus. Their medial borders are located on the medial wall of the hemisphere, coincident with the dorsal lip of the cingulate sulcus. Their lateral borders coincide with the medial lip of the suprasylvian sulcus and correspond architectonically – and physiologically – to the visual areas of this sulcus. The posterior border of PPc corresponds to the anterior border of area 18 (on the lateral gyrus) and area 21 (on the suprasylvian gyrus), anterior to the tip of the lateral sulcus (Innocenti *et al.*, 2002; Manger *et al.*, 2002). Anteriorly, PPc is bordered by PPr. The anterior border of PPr is with caudal somatosensory cortex (SIII, the area posterior to SI), <1 mm caudal to the junction of the ansate and lateral sulci.

Electrophysiological Mapping

Subdivision into Two Areas

Two extensive and separate representations of the visual field were found in the posterior parietal cortex, one in PPc the other in PPr. The posterior border of PPc abutted the anterior border of areas 18 and 21, and marked a reversal in the progression of the receptive fields around the relative periphery of the visual field [for a definition of the concept of relative periphery see Manger *et al.* (Manger *et al.*, 2002)]. Moreover, the size of the receptive fields increased markedly. The border between PPc and PPr was coincident with a reversal of receptive field progressions around the zero meridian. The size and shape of the receptive fields were not significantly altered; however, in PPr, we obtained responses to both visual and somatic stimulation. Both areas were bordered laterally by the suprasylvian visual region. This border corresponds to a reversal in receptive field progressions around the upper edge of the visual field. Moreover, in the suprasylvian regions the responses are stronger and the receptive fields are larger. The medial border of both areas is at the dorsal lip of the cingulate sulcus. Although we did not record in the medial wall of the hemisphere close to the midline of the hemisphere, the receptive fields were found in the lower portion of the visual field. Thus, it is reasonable to assume that the medial border of the posterior parietal fields represents the lower edge of the visual field. The anterior border of PPr was marked by the disappearance of responses to visual stimulation and a reversal in the progression of the somatic receptive fields around the periphery of the body, i.e. digit tips, ventral midline, and rostrum.

Visuotopic Representation in PPc

The representation of the visual field in PPc (Figs 1–3) appears to be relatively simple, unlike those previously found for areas

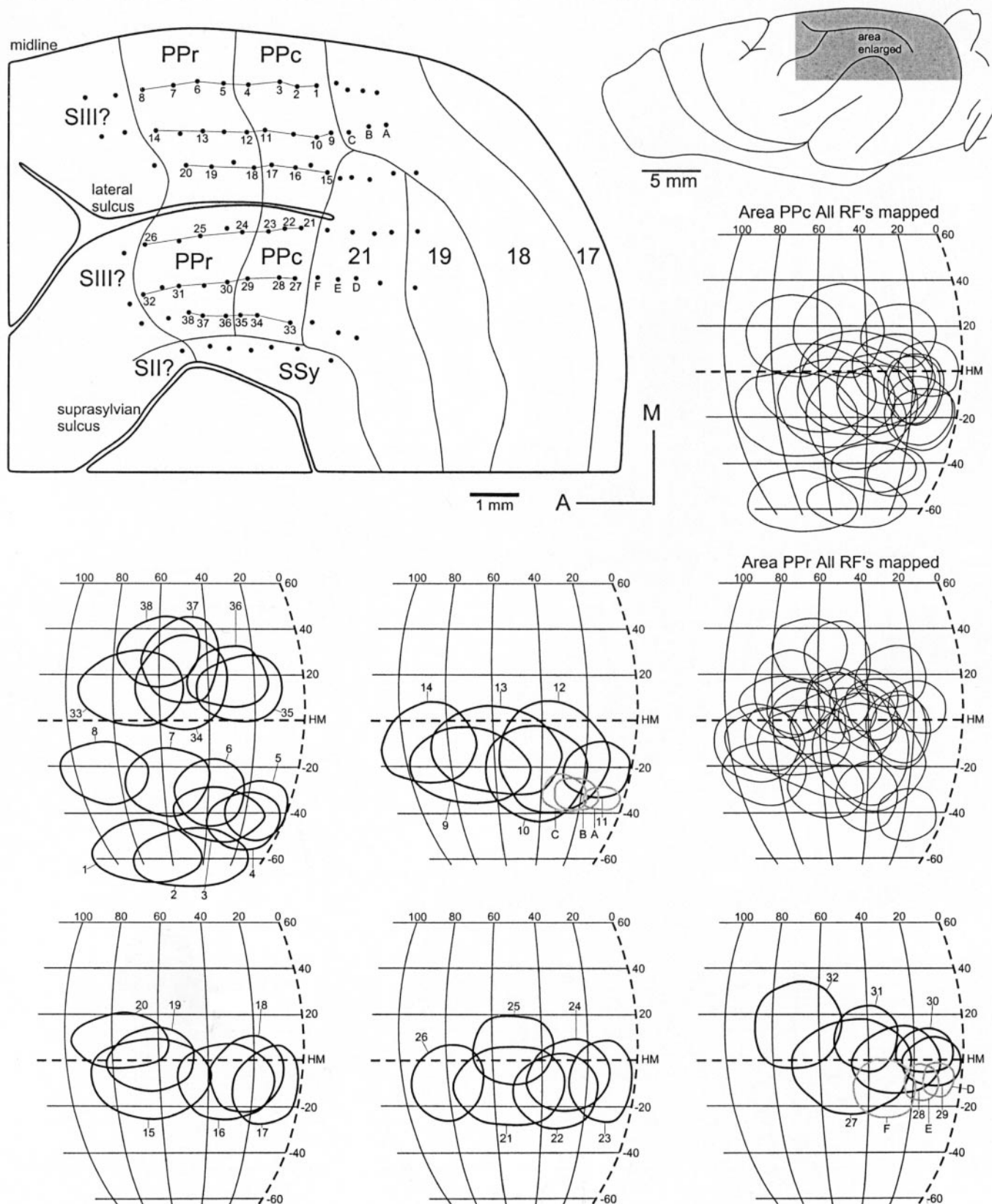


Figure 1. Reconstruction of the retinotopy within areas PPc (posterior parietal caudal area) and PPr (rostral area), with examples of receptive field progressions. Numbers at recording sites correspond to the illustrated receptive fields. A reversal in the progression of receptive fields around the zero meridian corresponds to the PPc/PPr boundary. Receptive fields in the lower portion of the visual field are found medially, on the lateral gyrus, while the upper visual field is found on the suprasylvian gyrus. The figurines to the right show all the receptive fields recorded within areas PPc and PPr in the present case. A, anterior; HM, horizontal meridian; M, medial; RF, receptive field; SII?, second somatosensory area (tentative); SSy, suprasylvian visual areas.

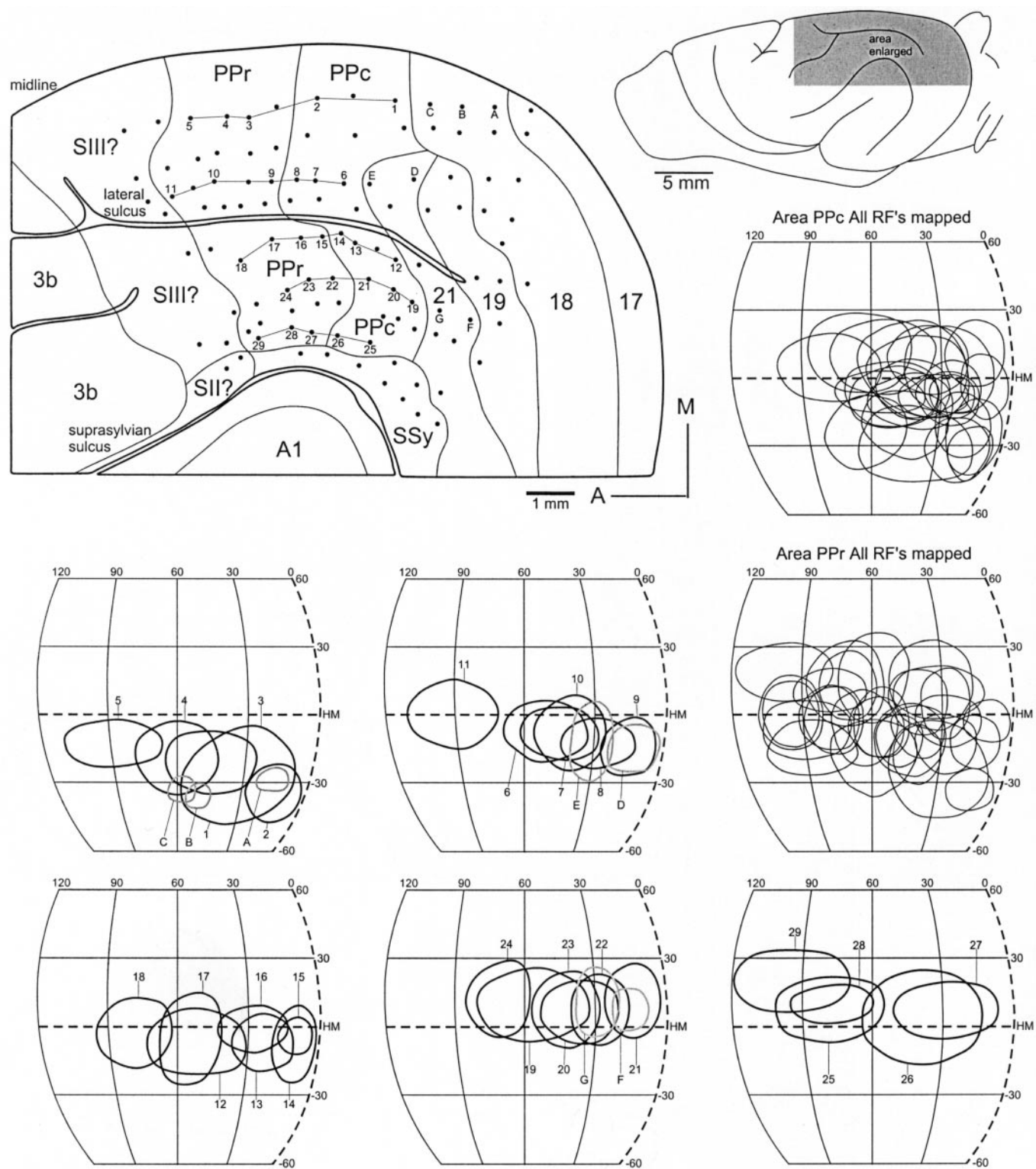


Figure 2. A second example of the retinotopy of areas PPr and PPc. Conventions as in Figure 2. A1, primary auditory cortex; 3b, primary somatosensory cortex.

18, 19 and 21 of the ferret (Manger *et al.*, 2002). The elevations show a smooth mediolateral progression, from $>40^\circ$ below the horizontal meridian in our most medial recording sites, to 30° or more above, at the border with the suprasylvian visual areas (Fig. 3). The lower hemi-field is found on the lateral gyrus, the upper hemi-field on the suprasylvian gyrus. The horizontal

meridian is represented in, and adjacent to, the lateral sulcus. The isoelevation lines are continuous with those found in areas 18 and 21, and run in an antero-posterior direction. The horizontal meridian representation is continuous with the dorsal horizontal meridian representation of area 21 (Manger *et al.*, 2002). The zero meridian is represented along the anterior

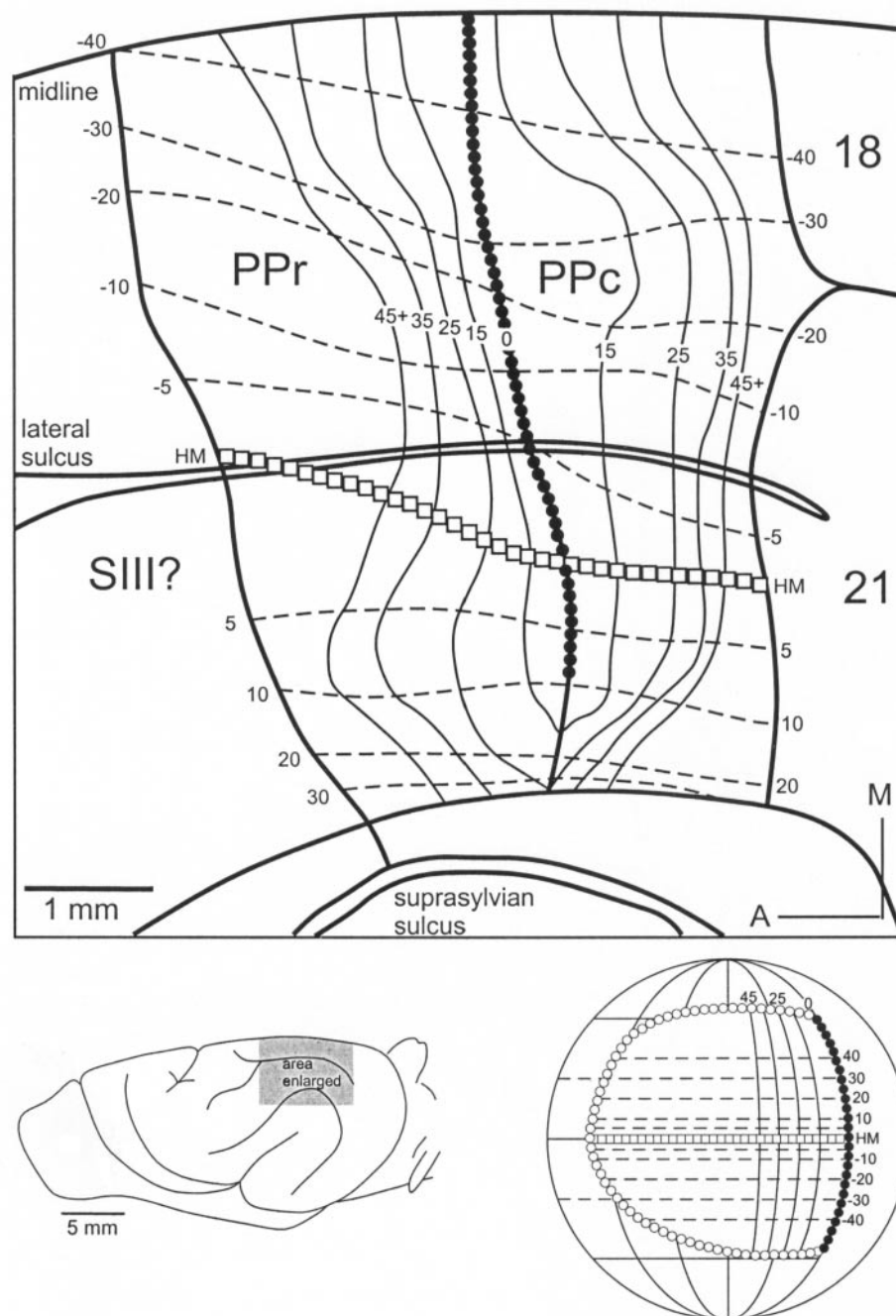


Figure 3. Summary diagram of the retinotopy within areas PPC and PPr. Isoelevation lines run antero-posteriorly and course, for the most part, parallel to the lateral sulcus. Isoazimuth lines run roughly parallel to the representation of the zero meridian at the PPC/PPr boundary. Numbers denote values of isoelevation (dashed lines) and isoazimuth (thin solid lines). Open squares represent the horizontal meridian (HM) and filled circles the zero meridian (0°; see globe figurine). This figure is an 'average' figure and is derived from several cases.

border of PPC; however, it stops ~1 mm short of its lateral border. The azimuths increase progressively and smoothly away from the zero meridian, up to and beyond 45° (Fig. 3). The isoazimuth lines generally run mediolaterally, parallel to the anterior border of the area; however, laterally, they curve forward and the line representing 25° of azimuth was found to cross the anterior border. In both areas an extensive portion of the visual field is represented (Fig. 4) but the lower field is magnified more than the upper field (Fig. 3).

The multi-unit receptive fields were significantly larger than those found in occipital visual areas – see Fig. 5 (Manger *et al.*,

2002). In general, they span between 30 and 45° in azimuth and between 20 and 30° in elevation. Thus, the majority of the receptive fields were elongated horizontally. There was a tendency for receptive field size to increase with eccentricity; however, this was not a statistically significant trend (Fig. 5). The neurons responded best to large stimuli, ~20° diameter, which were moved at a moderate speed across the receptive field. The strongest responses were evoked by 'sweeping' the stimulus across the receptive field, regardless of direction; these stimuli evoked reliable responses with stable receptive field centers and borders within the time of observation (3–5 min).

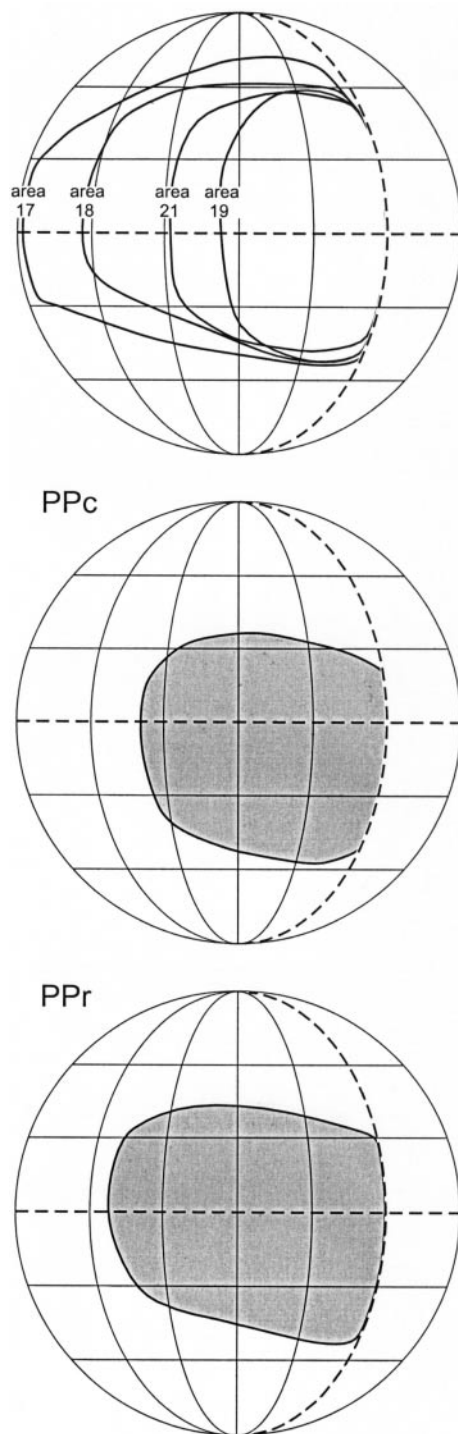


Figure 4. Extent of visual field representation, with each area compared to areas 17, 18, 19 and 21 (Law *et al.*, 1988; Manger *et al.*, 2002). Areas PPC and PPr represent approximately the same amount of the visual field, both with a bias towards the lower visual field. Meridians are at 30° from each other.

Visuotopic Representation in PPr

The representation of the visual field in PPr was an approximate mirror reversal of that seen in PPC, with a similar simple topography (Figs 1–3). Elevations show a smooth progression from ~40° below the horizontal meridian to 30° above, as recordings advance from medial to lateral (Fig. 3). The lower hemi-field is found on the lateral gyrus and the upper hemi-

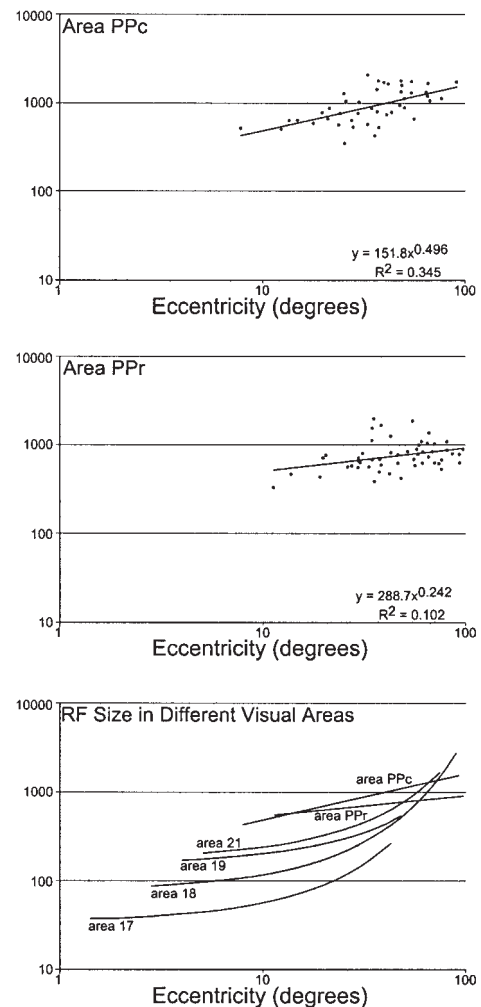


Figure 5. Plots of multi-unit receptive field size (degrees²) against eccentricity. Eccentricity is given as the distance (in degrees) of the center of the receptive field from the intersection of the zero and horizontal meridians. The data were plotted in Excel and an equation describing the line of best fit and its correlation coefficient (R^2) generated. In both areas the receptive field sizes are large but comparable and there is a weak tendency for the receptive fields to increase in size with increasing eccentricity. Receptive fields in both PPC and PPr are significantly larger than those in the occipital visual areas of the ferret (Manger *et al.*, 2002).

field on the suprasylvian gyrus. The horizontal meridian is represented in and around the lateral sulcus. The isoelevation lines are continuous with those of PPC and course in an antero-posterior direction. The zero meridian is represented at the posterior border of this area, co-extensive with that of PPC and, similarly, does not extend to the lateral edge of the field. Azimuths progress smoothly away from the zero meridian, up to and beyond 45° (Fig. 3). The isoazimuth lines course parallel to the posterior border of the area, except at the lateral edge of the field, where they curve backward to meet the forward-coursing isoazimuth lines of PPC.

The size of the receptive fields in PPr was similar to those of PPC, with a weak tendency to increase with eccentricity (Fig. 5). Many receptive fields were elongated horizontally and the neurons were most responsive to large stimuli moved at a moderate speed through the receptive field. The responses elicited by these stimuli were consistent and provided reliable and stable receptive field centers and borders (within the time observed).

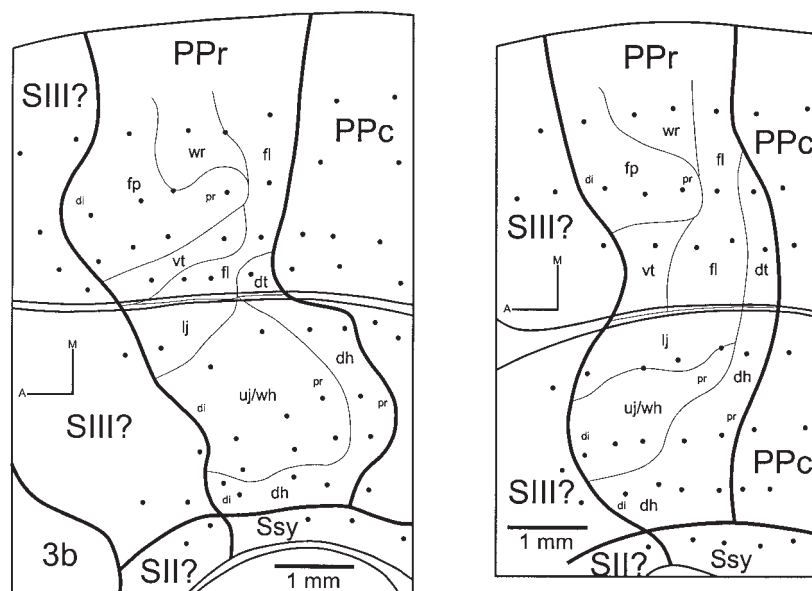


Figure 6. Reconstructions of the somatotopic map found in area PPr in two cases. The representation of the head is found on the suprasylvian gyrus and that of the post-cranial body on the lateral gyrus. The rostrum and digit tips are found anterior in the field, while the dorsal midline is located at the posterior border of the field. 3b, primary somatosensory cortex; A, anterior; di, distal; dh, dorsal head; dt, dorsal trunk; fl, forelimb; fp, forepaw; lj, lower jaw; M, medial; pr, proximal; Ssy, suprasylvian visual areas; SII?, second somatosensory area; SIII?, third somatosensory area (tentative); uj/wh, upper jaw/whiskers; vt, ventral trunk; wr, wrist.

Somatotopic Representation in PPr

A representation of the contralateral body was found within this cortical area (Fig. 6). The organization is not as topographically precise as in primary somatosensory cortex; however, similarities in the organization of the map were seen across all cases. The post-cranial body was represented on the lateral gyrus and the head on the suprasylvian gyrus. Thus, the mediolateral sequence of body parts, typical for parietal somatosensory areas, was found. At the posterior edge of PPr, the representation of the dorsal body, including the shoulder, dorsal trunk and dorsal head, was found. As recordings were made progressively anterior, the receptive fields were localized distally on the forelimb, on the ventral trunk, or the rostrum. Expanded representations of the forepaw, on the lateral gyrus, and upper jaw/vibrissae, on the suprasylvian gyrus, occupied the majority of the field.

The best responses were evoked by passive motion (especially for the forelimb), or light tapping (for whiskers and upper lip). Often the responses would habituate following repeated stimulation. The receptive fields were large and often encompassed a significant portion of the body, for example the entire forelimb, or the entire vibrissal pad. When recordings were made anterior to PPr, in caudal somatosensory regions (Hunt *et al.*, 2000), the size of the somatic receptive field decreased dramatically and the neurons responded to light cutaneous stimulation. Additionally, the progression of receptive fields reversed from the digit tips and ventral body, towards more proximal and dorsal regions of the body.

Architectonic Analysis

The physiological representations in posterior parietal cortex lie in a territory architectonically distinct from the surrounding visual (occipital and suprasylvian), somatosensory (SII) and cingulate areas. Within posterior parietal cortex, the architectonic differences are sometime subtle; however, these differences are for the most part compatible with the electrophysiological maps.

Cytoarchitecture

With Nissl staining, two cytoarchitectonic fields were determined. The first field extended from the dorsal lip of the cingulate sulcus, across the lateral gyrus to the fundus of the lateral sulcus and corresponds to the representation of the lower visual field in both areas PPc and PPr (see above). All six cortical layers could be clearly differentiated (Figs 7B,D and 8A) and show the relative thicknesses typically described for posterior parietal cortex of the macaque monkey (Jones *et al.*, 1978; Pandya and Seltzer, 1982). Layer 1 is thick in comparison to the occipital visual areas. Numerous granule cells make up a thin layer 2, which has a slightly blurred boundary with the pyramidal cells of layer 3. No sublamination of layer 3 is evident and layer 4 is thick, with densely packed granule cells. Layer 5 exhibits a relatively low cell density and a low number of large pyramidal cells. Loosely arranged palisades, ranging in width from one to four cells, characterize layer 6. The cytoarchitectural features are not affected by the presence of the lateral sulcus and medial wall of the hemisphere, except for specific compressions and expansions of layer thickness (Welker, 1990); further subdivision of this field appear unwarranted.

The second cytoarchitectonic field extends lateral to the first, from the fundus of the lateral sulcus, across the crown of the suprasylvian gyrus to the medial lip of the suprasylvian sulcus and corresponds to the physiologically defined upper visual field representation of PPc and PPr (see above). Two features distinguish this field from the more medial one (Figs 7A,C and 8A). The first is a clear sublamination of layer 3, with many large pyramidal cells in the deeper part of layer 3, forming a layer 3c. The second is the presence of several larger pyramidal cells throughout layer 5. Again, except for specific compressions and expansions of layers due to the sulcal pattern, there appeared to be no cytoarchitectural differences that warranted subdivision.

Myeloarchitecture

The entire posterior parietal cortex of the ferret was moderately myelinated, less so than area 18, but more than areas 19 and 21

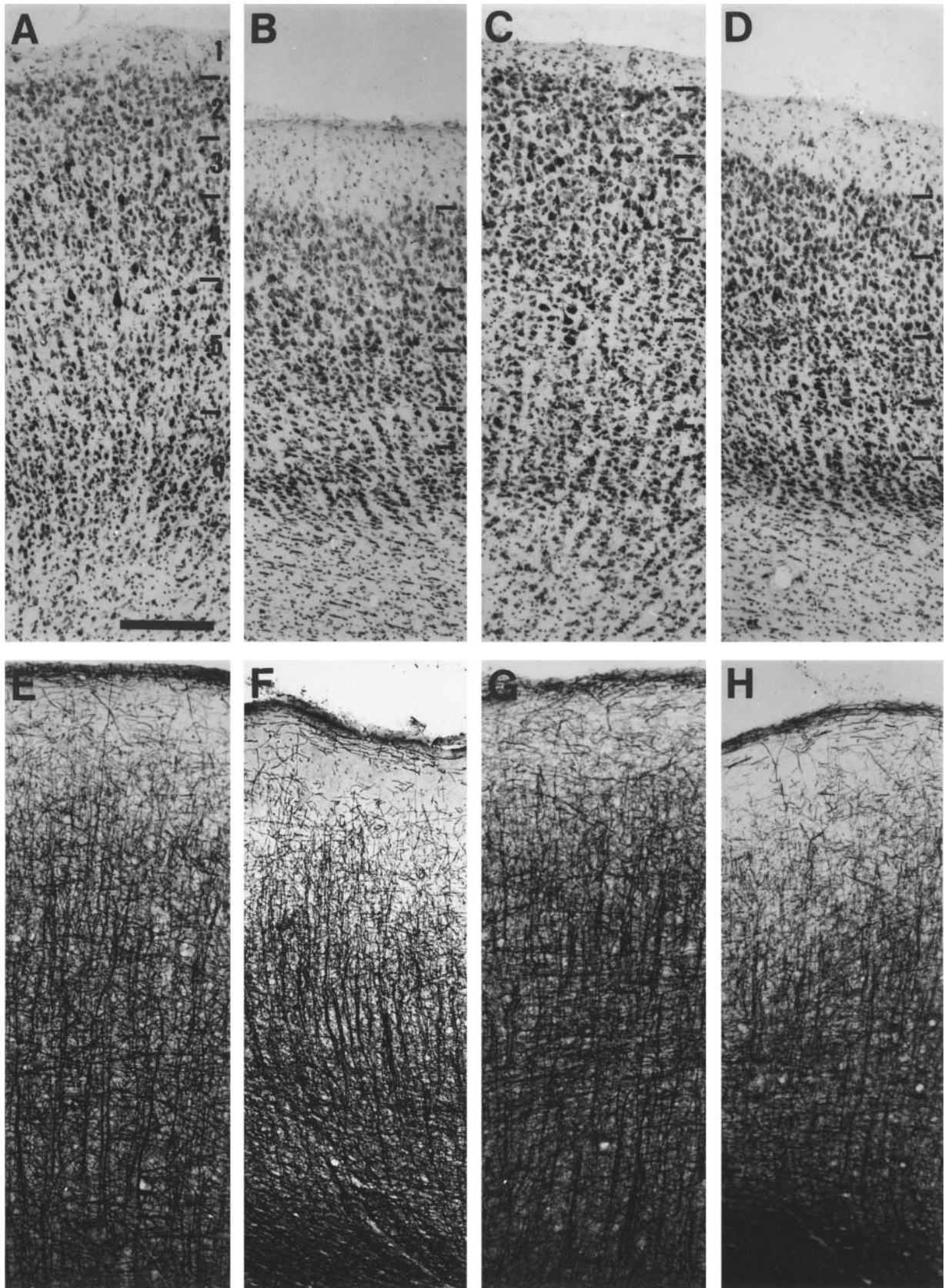


Figure 7. Architecture of areas PPc (A, B, E, F) and PPr (C, D, G, H). Photomicrographs from sections stained with cresyl violet (A–D) or myelin (E–H). (A) and (E) are from area PPc on the suprasylvian gyrus, while (B) and (F) area from area PPc on the lateral gyrus. (C) and (G) are from suprasylvian gyrus and (D) and (H) from lateral gyrus in area PPr. Note the lack of cytoarchitectural differences between (A) and (C), and (B) and (D), contrasted with the differences between (A) and (B), and (C) and (D). (E) and (F) show similarities in the radial fascicle morphology, as do (G) and (H); however, the differences between these two are striking. Moreover, within (G) and (H) there is a marked increase in tangential fibre density in layer 4. Scale bar = 200 μ m.

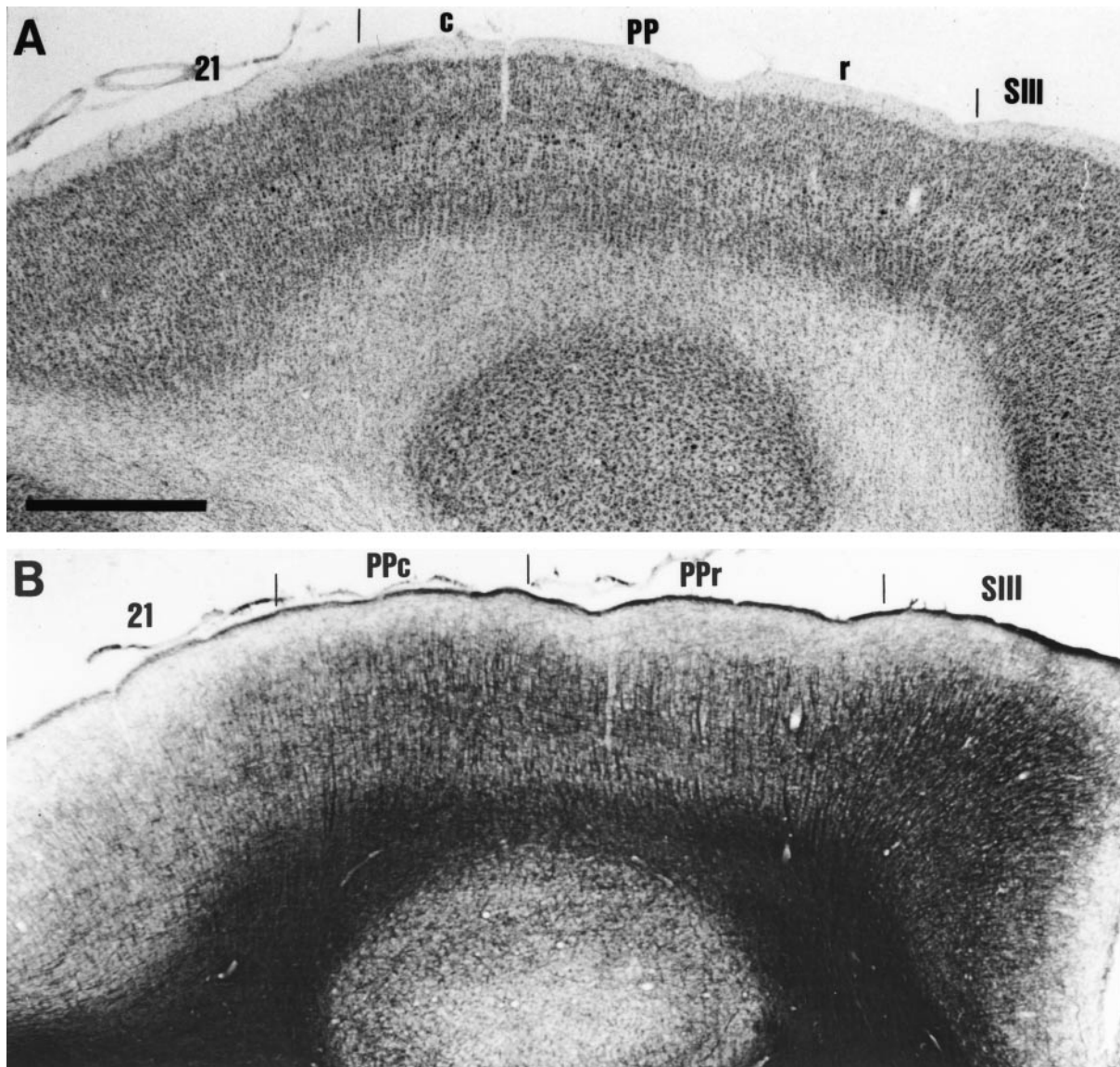


Figure 8. Architecture of areas PPc and PPr and surrounding regions. Low-power photomicrographs of adjacent parasagittal sections through the suprasylvian gyrus stained for Nissl (A) and myelin (B). In the Nissl-stained section, it can be seen that the thickness of layer 4 in posterior parietal cortex distinguishes this region from area 21 posteriorly and area SIII anteriorly. In the myelin-stained section, posterior parietal cortex is differentiated from area 21 by increased myelin density and from area SIII by lower myelin density. Areas PPc and PPr can also be distinguished by their individual features (see Fig. 7). Scale bar = 1 mm for both photomicrographs.

(Innocenti *et al.*, 2002). Four myeloarchitectonic fields could be discerned based on the configuration of radially and tangentially oriented fibers.

The first myeloarchitectonic field coincided with the lower visual field representation of area PPc (see above) and was located immediately anterior to area 18, from the dorsal lip of the cingulate sulcus to the fundus of the lateral sulcus. Within this field, numerous loosely arranged radial fascicles could be seen traversing the cortical layers, uninterrupted, from layer 6 to layer 2. A moderate density of tangential fibers was evident in layer 5 and a light density in layer 3 (Figs 7F and 8B).

The second myeloarchitectonic field was coincident with the upper visual field representation of area PPc (see below), located immediately anterior to area 21, from the fundus of the lateral sulcus to the medial lip of the suprasylvian sulcus. This field exhibited the same density of loosely arranged, but uninterrupted, radial fascicles running from layer 6 to layer 2 as the first

field; however, the density of tangential fibers in layers 3, 5 and upper layer 6 were appreciably higher (Figs 7E and 8B). This increase of tangential fibers is coextensive with the increase in large pyramidal cells in layers 3, 5 and upper 6 of this region (see above).

A third myeloarchitectonic field was located anterior to the first, corresponded to the lower visual field representation in area PPr (see above) and extended from the dorsal lip of the cingulate sulcus to the fundus of the lateral sulcus. The density of radial fascicles in this field was comparable to the others described above; however, the fascicles were rather short, spanning a maximum of two cortical layers. Fascicles in this field were often limited to either layers 2 and 3 or 5 and 6, but generally avoided layer 4. A moderate to low density of tangentially oriented fibers was found from layers 2 to 6, but with a significant increase in fiber density in layer 4 (Figs 7H and 8B).

The fourth myeloarchitectonic field was located anterior to the second, corresponded to the upper visual field representation of area PPr (see above) and extended from the fundus of the lateral sulcus to the medial lip of the suprasylvian sulcus. The density and structure of the radial fascicles are similar to those seen in the third myeloarchitectonic field; however, the density of tangentially oriented fibers is greater, especially so in layers 3, 5 and upper layer 6 (Figs 7G and 8B). This again is coextensive with the increased number of large pyramidal cells in this cortical region (see above). Similar to the third field, the greatest density of tangential fibers occurs in layer 4.

Connectivity of the Posterior Parietal Cortical Areas PPr and PPr

Thalamocortical Connections

The ferret visual thalamus has not been fully defined in any previous studies; however, the laminated LGN has been described in detail (Linden *et al.*, 1981; Zahs and Stryker, 1985). Herbert (Herbert, 1963), described a lateralis posterior nucleus (LP) and our own ongoing studies (unpublished results) indicate the existence of an architectonically distinct pulvinar, which, as in the cat, contributes to an LP-pulvinar complex. LP can be subdivided into three regions – LPI (lateral), LPI (interjacent) and LPM (medial) – as described for the cat (Graybiel and Berson, 1980; Raczkowski and Rosenquist, 1983; Updyke, 1981, 1983). However, the LP-pulvinar complex is rotated relative to the cat, in a manner similar to that described for the LGN (Zahs and Stryker, 1985), the pulvinar being located anteriorly to LP in the ferret.

Area PPr. For injections localized in upper or lower field representations of this area, the pattern of retrograde labeling in the thalamus was similar in terms of nuclear subdivisions; however, differences related to topographic representations were noted (Fig. 9). In all cases, labeled neurons were found throughout the C-lamina of the LGN, but the location changed as predicted by the retinotopic organization of this lamina (Zahs and Stryker, 1985). Neurons labeled from injections in upper or lower visual field representations showed some overlap in their distribution; however, in general, neurons labeled from lower field injections were found ventral to those labeled from upper field injections. The projections from LP provide the strongest thalamic input to this area. A loose sheet of labeled neurons stretched throughout the ventral half of LP and continued into the pulvinar. Within LP, neurons labeled from upper and lower field injections overlapped extensively; however, in general, neurons labeled from lower visual field injections were found ventral to those labeled from upper field injections. Within the pulvinar, neurons labeled from upper and lower field injections were more extensively intermingled; however, there was a rough ventro-posterior lower-field and dorso-anterior upper-field correspondence. The most medial portion of the pulvinar remained free of labeled neurons; however, they were found throughout the antero-posterior and dorso-ventral extent of the lateral portion. Labeled neurons topographically intermingled, as observed in the pulvinar, were seen in the ventralis anterior (VA) nucleus. These were restricted to the most dorsal portion of this nucleus, adjacent to the pulvinar.

Area PPr. Injections localized in upper or lower field representations of this area exhibited similar patterns of retrograde labeling in thalamic nuclei with some minor distinctions with regard to topography (Fig. 10). No region of the LGN was labeled

in any of the cases studied. A small number of labeled neurons was found within the anterior one-third of LP. These showed a topographic consistency as seen for area PPr, with lower visual field injections labeling neurons ventrally in LP. The densest region of retrograde labeling was located within the pulvinar. Here, there was some evidence of a topographic relationship; however, there was also extensive intermingling of cells projecting to upper and lower field injection sites. Labeled neurons were observed throughout the entire extent of the pulvinar, including its medial portion, indicating that the pulvinar of the ferret may in fact be divisible into two distinct regions. Labeled neurons were also found in the dorsal portion of VA. Here, the density of labelling was almost comparable to that seen in the pulvinar and extended just beyond the anterior limit of this nucleus. Some weak topography in the distribution of labeled cells in VA was observed.

Callosal Connections

The overall pattern of callosal connections within the posterior parietal cortex was examined to determine if, as for the occipital visual areas of the ferret (Innocenti *et al.*, 2002; Manger *et al.*, 2002), it might provide additional criteria for the identification of areal borders. Multiple injections were centered on the posterior parietal cortex, the spread of tracer covering these areas plus the medial parts of areas 17, 18 and 21. The pattern of callosal connections in the contralateral posterior parietal areas was relatively homogeneous (Fig. 11). Unlike in the occipital visual areas (Innocenti *et al.*, 2002), neither callosal bands nor local changes in the density of labeled neurons corresponding to the representation of the zero meridian or periphery of the visual hemifield were identified. Instead, the density of labelled cells was lower on the lateral gyrus than the suprasylvian gyrus (lateral gyrus density ~70% of the suprasylvian gyrus), indicating stronger connections of the upper visual field and head representations.

Callosally projecting neurons were also found in the extrastriate visual areas (areas 18, 19, 21, the temporal and suprasylvian areas) and the somatosensory area immediately anterior to the two parietal areas studied here, the region termed SIII, or area 5, in the carnivore (Johnson, 1990). The anterior border of this somatosensory area, its border with area 3b (SI, or primary somatosensory cortex), marks a sharp drop in the number of callosally projecting neurons. Within area 3b (identified by dense CO reactivity and exploratory physiological recordings), only very few callosally labeled neurons were ever found, mostly at its posterior border. However, a small cluster of labeled neurons was consistently found in a region of 3b (Fig. 11). Thus, two aspects of callosal connectivity emerge: first, the distribution of neurons is relatively homogenous and, secondly, there are extensive callosal projections from area SIII, but not area 3b.

Supragranular and infragranular laminae of callosally projecting neurons were readily identifiable throughout both PPr and PPr (Fig. 12). The supragranular lamina consisted mainly of layer 3 neurons, with occasional neurons in layers 2 and 4. The infragranular lamina was located in layer 6, with occasional neurons in layer 5. The majority of the callosally projecting neurons were small to medium-sized pyramids; however, there were several significantly larger pyramidal neurons as well, i.e. with soma areas >90 μm^2 (Fig. 13). The percentage of large callosally projecting neurons in the supragranular lamina was smaller in the lateral gyrus (~8%) than in the suprasylvian gyrus (~18%). A similar difference was found for the infragranular lamina (2% lateral gyrus, 11% suprasylvian gyrus). The differences in soma size between callosally projecting neurons in the

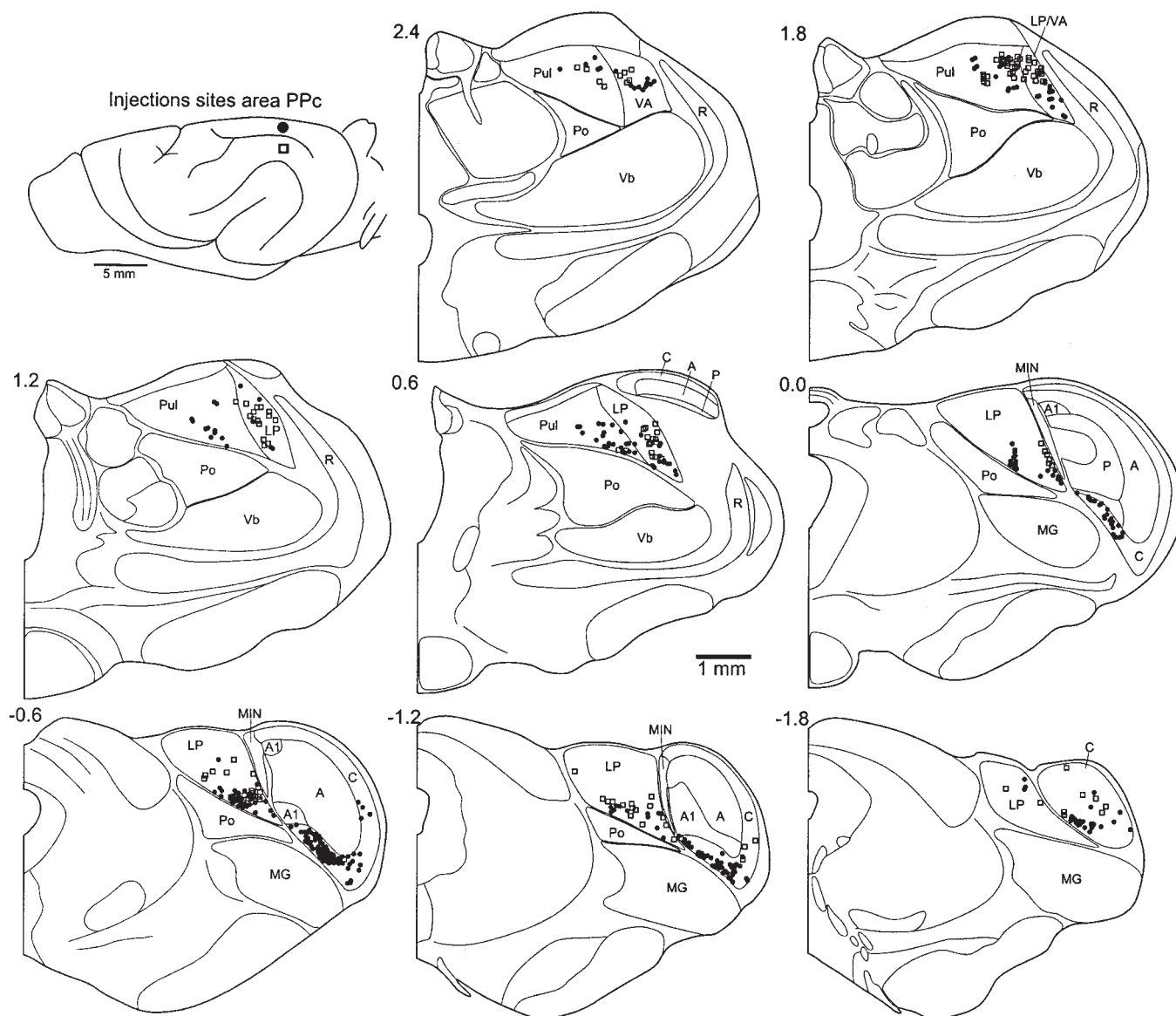


Figure 9. The location of retrogradely labeled neurons in the thalamus of the ferret following injections in the upper (open squares) and lower (closed circles) field representations in area PPc (posterior parietal caudal area). Labeled neurons were found throughout the C lamina of the LGN and the lateral posterior nucleus (LP), and in the medial portion of the pulvinar (Pul) and the caudal portion of the ventral anterior nucleus (VA). Numbers next to each thalamic section figurine represent the approximate stereotaxic location in millimeters anterior or posterior to stereotaxic zero. A, A lamina of LGN; A1, A1 lamina of LGN; LGN, lateral geniculate nucleus; MIN, medial interlaminar nucleus; MG, medial geniculate nucleus; Po, posterior nucleus; R, reticular nucleus; Vb, ventrobasal complex.

lateral and suprasylvian gyri were quantified on a sample of 400 cells within six randomly chosen fields in the two areas, near their common border. The differences in the distribution of cell sizes were statistically significant ($P < 0.05$ for supragranular layers; $P < 0.001$ for infragranular layers; non-parametric median rank test, as implemented in JMP software, SAS Institute). This differential distribution of callosally projecting neurons with large soma size appears directly related to the cytoarchitectonic differences between the gyri (see earlier results and Fig. 7), i.e. to the differences in cell sizes in layer 3. Thus, the differential callosal connectivity may be the basis for the difference in cytoarchitecture and is accompanied by the increased horizontal fibre density in the corresponding layers of the suprasylvian gyrus as compared to the lateral gyrus (see earlier results and Fig. 7).

Discussion

The Parcellation of Posterior Parietal Cortex in the Ferret

We have delineated two areas in the posterior parietal cortex of the ferret – PPc and PPr (Fig. 14). These areas are bordered posteriorly by visual areas 18 (on the lateral gyrus) and 21 (on the suprasylvian gyrus), anteriorly by somatosensory area SIII, laterally by suprasylvian visual areas and medially by the cingulate sulcus. Each area contains an independent and near complete representation of the visual field, nearly as extensive as that of area 18 and somewhat larger than that of areas 19 and 21 (Manger *et al.*, 2002), with a bias towards the representation of the lower visual field.

Electrophysiological observations provided compelling

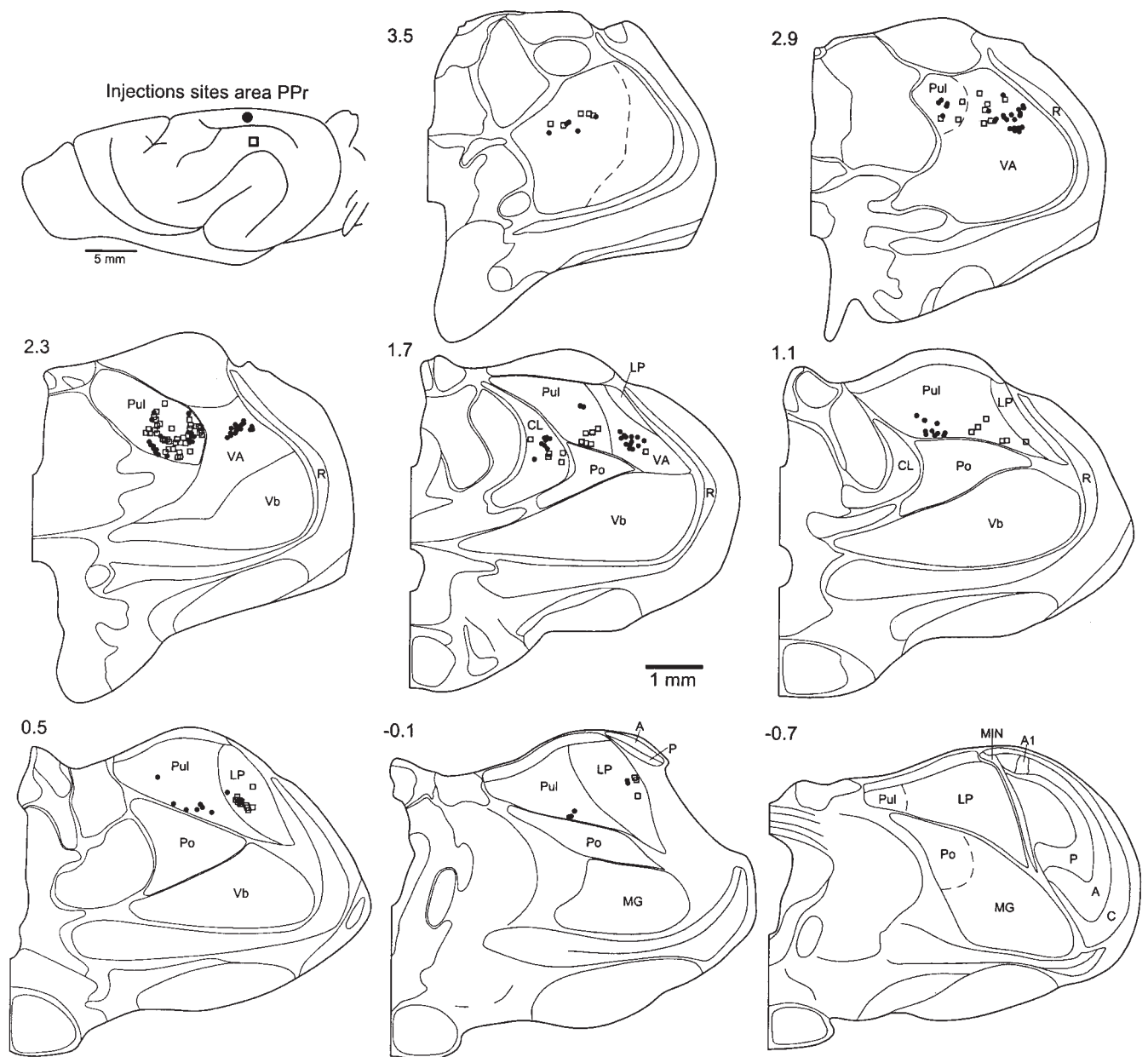


Figure 10. The location of retrogradely labeled neurons in the thalamus of the ferret following injections in the upper (open squares) and lower (closed circles) field representations in area PPr (posterior parietal rostral area). Conventions and abbreviations as in Figure 8. Labeled neurons were found in the most rostral portion of the lateral posterior nucleus (LP), throughout both medial and lateral portions of the pulvinar (Pul) and into the ventral anterior nucleus (VA). Unlike area PPc, area PPr does not receive projections from the lateral geniculate nucleus.

criteria for subdividing posterior parietal cortex into two, rostrocaudally distinct, mediolaterally oriented, areas. These criteria include the mediolateral continuity in the retinotopic representations in each area (from low elevations medial to high elevations lateral), the duplication of retinal representation between the two areas (with a reversal around the zero meridian at their mutual border) and, finally, the responsiveness of PPr, but not PPc, to somatosensory stimuli. The somatic representation confirms the mediolateral continuity of PPr, with the postcranial body being represented medially, on the lateral gyrus, and the head laterally, on the suprasylvian gyrus. The electrophysiological parcellation is supported by two myeloarchitectonic features. Within PPc, the radial fascicles spanned, uninterrupted, from deep to superficial layers, in both the upper

and lower visual field representations. Throughout PPr, the radial fascicles exhibited a discontinuous morphology and often spanned only two cortical layers. Additionally, area PPr exhibited a high density of tangential fibres in layer 4, which was not seen in PPc.

The distribution of retrogradely labeled thalamocortical neurons is also parsimonious with the electrophysiological parcellation. Irrespective of their retinotopic locations, tracer injections in PPc resulted in labeled neurons in similar thalamic nuclei, including the C-lamina of the LGN, LP and pulvinar. In contrast, injections centered in PPr labeled LP and pulvinar, but never the LGN. Thus, different groups of thalamic nuclei project to each physiologically defined area, although the same complement of nuclei project to upper and lower visual field

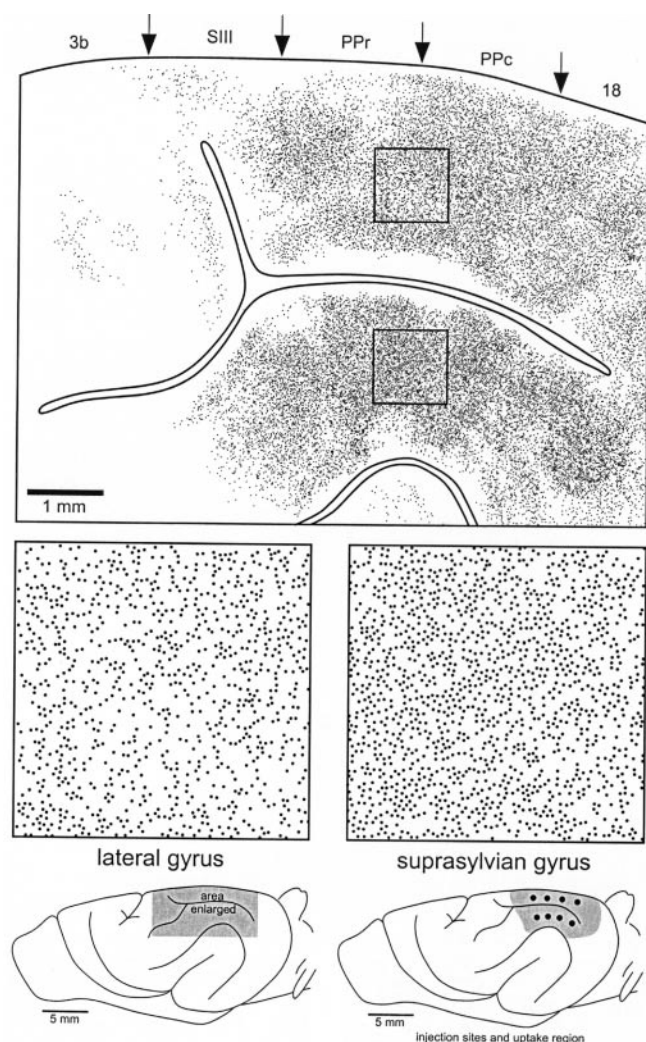


Figure 11. Distribution of callosally projecting neurons in a semi-flattened section of the parietal cortex of the ferret. Each labeled neuron is represented by a dot. The distribution is fairly homogeneous across posterior parietal cortex; however, the upper visual field representation (on the suprasylvian gyrus) exhibits a higher density of callosal neurons than the lower visual field representation (on the lateral gyrus). No correspondence of callosally projecting neurons to the zero meridian was found. There is a marked drop in the number of labeled neurons at the posterior border of primary somatosensory cortex; however, the third somatosensory area (SIII) contains many neurons. Approximate area boundaries are marked. The figurines at the bottom of the figure represent the area enlarged and the injection sites and uptake region from the opposite hemisphere.

representations within each area. The pattern of retrograde labeling also validates the retinotopy found physiologically, with upper visual field injections labeling neurons dorsally and lower field injections labeling neurons ventrally within the thalamic nuclei (Zahs and Stryker, 1985). Thus, electrophysiological mapping, radial fascicle morphology, layer 4 tangential fiber density and thalamocortical projections indicate that the posterior parietal cortex of the ferret should be subdivided into two rostrocaudally distinct, mediolaterally oriented, areas.

Unlike in occipital areas (Innocenti *et al.*, 2002; Manger *et al.*, 2002), the correspondence between the electrophysiological maps and the cytoarchitectonic fields of the posterior parietal cortex is not simple. The cytoarchitecture clearly distinguishes posterior parietal cortex from the surrounding visual (occipital

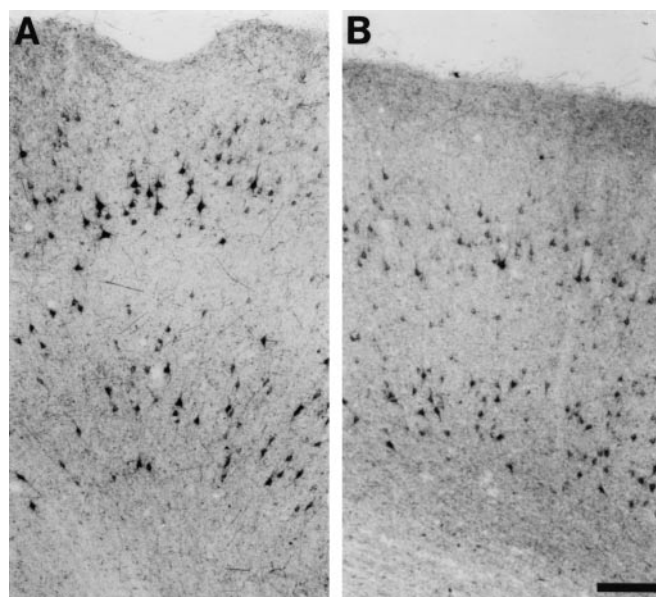


Figure 12. Photomicrographs of callosally projecting neurons labeled with WGA-HRP at the border of PPc and PPr. (A) is taken on the suprasylvian gyrus, (B) on the lateral gyrus. Note the numerous larger pyramidal neurons in the supragranular layers and the somewhat larger cells in the infragranular layers of (A) as compared to (B). Scale bar = 100 µm.

and suprasylvian), somatic (SIII) and cingulate cortical regions. The medial cytoarchitectonic field – termed area 7 in the kinkajou (Brodmann, 1909) – corresponds to the lower visual field representations of both areas and the lateral cytoarchitectonic field (termed area 52 in the kinkajou) corresponds to the upper visual field representations. The differences in the two cytoarchitectonic fields may relate to callosal connectivity. Compared to the medial cytoarchitectonic field, the lateral field possessed a distinct layer 3c that contained numerous large pyramidal cells and also had larger and a slightly higher density of callosally projecting cells. These callosally based differences are paralleled by the increased tangential fiber density in the suprasylvian gyrus as compared to the lateral gyrus. Thus, while a different scheme of subdivision of the ferret parietal cortex might be reached using cytoarchitectonics – as exemplified for the kinkajou (Brodmann, 1909) – these differences appear to relate to callosal connectivity and might have a specific functional basis for processing within the cortical areas (see below). Hence, we adopt the parcellation indicated by electrophysiological mapping, thalamocortical connectivity and aspects of the myeloarchitecture.

Posterior Parietal Cortex in Other Species

All eutherian mammals possess multi-modal cortex between the unimodal visual and somatosensory areas (Brodmann, 1909; Kaas, 1995; Krubitzer, 1995). The extent to which this multi-modal, or association, cortex is elaborated in different lineages varies greatly, from a small strip in, for example, rodents, to a complex array in primates (Johnson, 1990). However, analyses of the organization of this cortical territory are few and, apart from the present study, data are essentially only available for the cat, flying fox and macaque monkey.

In the cat, several architectonic studies present a variety of schemes for this region of cortex, none of which match the present subdivisions in the ferret (Gurewitsch and Chatschaturian, 1928; Hassler and Muhs-Clement, 1964; Heath and Jones, 1971;

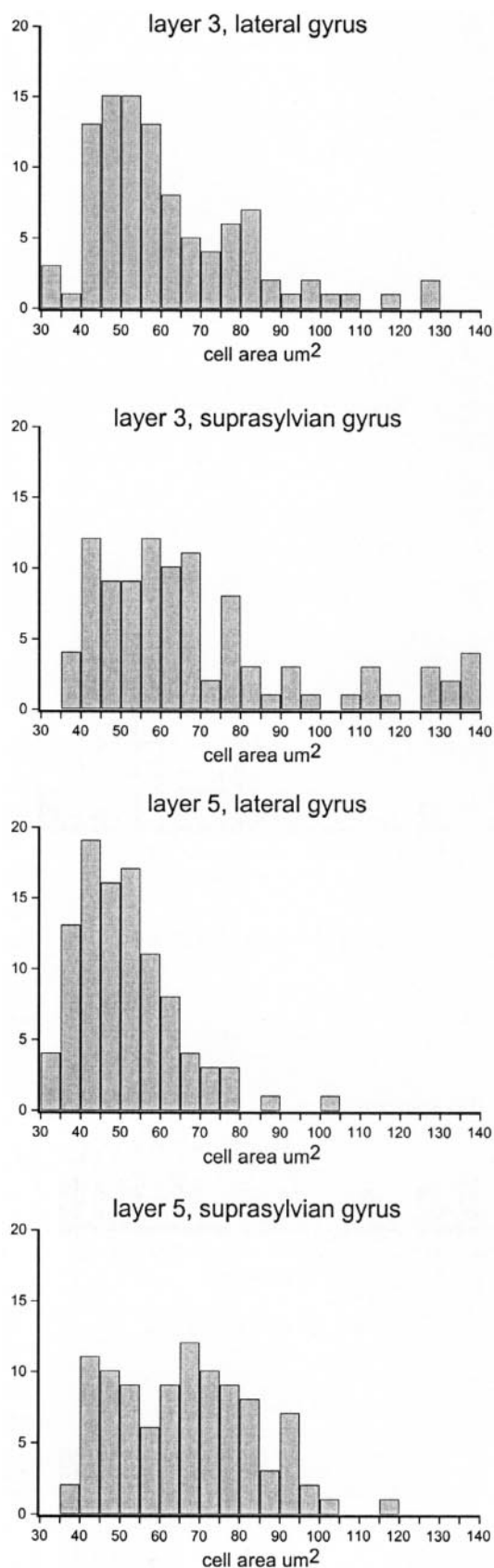


Figure 13. Frequency histograms of the somatal areas of callosally projecting neurons in the posterior parietal cortex. These histograms quantify the observable differences shown in Figure 11. An interesting feature is that the clear differences from direct observation (Fig. 12) amount to quantification differences of only ~10% (see text).

Sanides and Hoffmann, 1969). However, it should be noted that in the kinkajou, Brodmann (Brodmann, 1909) identified two cytoarchitectonic fields in this region which match those found for the ferret in the present study. Recently, a lower visual field representation was found anterior to area 21, on the suprasylvian gyrus of the cat (Pigarev and Rodinova, 1998), which possibly corresponds to part of area PPC in the ferret. Behavioral studies, with reversible cooling deactivation of the same region in the cat, failed to demonstrate any functional deficits (Lomber, 2001). Thus, the present study in the ferret provides the most complete description of this region for a carnivore.

In the flying fox, immediately anterior to the medial portions of V2 and V3, a topographically organized unimodal visual area was found – area OP, the occipitoparietal area (Rosa, 1999) – with a complete representation of the visual field, large receptive fields, coarse topography and a magnified representation of the lower visual field. This area appears similar to area PPC in the ferret. Anterior to OP, Rosa (Rosa, 1999) defined a multi-modal area termed PP (the posterior parietal area), where neural clusters responded to both visual and somatic stimuli; however, no topography was found. This region appears to be similar to PPr in the ferret. Thus, in the flying fox and the ferret, the posterior parietal cortex appears to consist of a caudal, unimodal visual area and a rostral, multi-modal area. These two areas in the flying fox and ferret might be direct homologues.

In a broader sense, it seems likely that PPC and PPr of the ferret correlate, as a whole, to the complex of posterior parietal areas described for the macaque within Brodmann's areas 5 and 7 (Seltzer and Pandya, 1980; Pandya and Seltzer, 1982; Baizer *et al.*, 1991; Geyer *et al.*, 2000; Lewis and Van Essen, 2000a,b). At this point, no direct area-to-area comparisons can be made between the ferret and macaque monkey; however, there are four similarities in organizational principles which indicate that the complex organization of parietal cortex in monkey may have evolved from a simpler organization such as that described here for the ferret. First, one pivotal cytoarchitectonic feature of the macaque is that the major difference noted between areas on the superior and inferior parietal lobules is the existence of a clear layer 3c in the latter (Seltzer and Pandya, 1980; Pandya and Seltzer, 1982). This difference is identical to that which we observed in the ferret between the lateral and suprasylvian gyri. Secondly, this cytoarchitectural difference is paralleled by the differential density of callosal connectivity (Ebner, 1969; Pandya and Vignolo, 1969), with the macaque superior parietal lobule having somewhat less dense connections than the inferior parietal lobule. We have shown that this difference in callosal connectivity appears to explain the cytoarchitectonic difference between lateral and suprasylvian gyri in the ferret, i.e. a clear layer 3c in the suprasylvian gyrus. Thirdly, as described here for the ferret, thalamic projection zones of some subdivisions of the macaque LP-pulvinar complex form mediolaterally oriented bands spanning the parietal cortex, including the intraparietal sulcus (Jones, 1985). Fourthly, both areas described in the ferret had a bias towards representation of the lower visual field. This bias was also observed in area OP of the flying fox (Rosa, 1999). The areas of the dorsal stream in the macaque monkey, and other primates, exhibit a similar bias in the representation of the visual field (Desimone and Ungerlieder, 1986; Previc, 1990; Galletti *et al.*, 1999a,b).

A greater complexity of parietal areas in the monkey as compared to the ferret would not be surprising, given the exceptional manual dexterity, eye movements and hand-eye coordination of the macaque monkey. However, based on the four similarities mentioned above, the most parsimonious

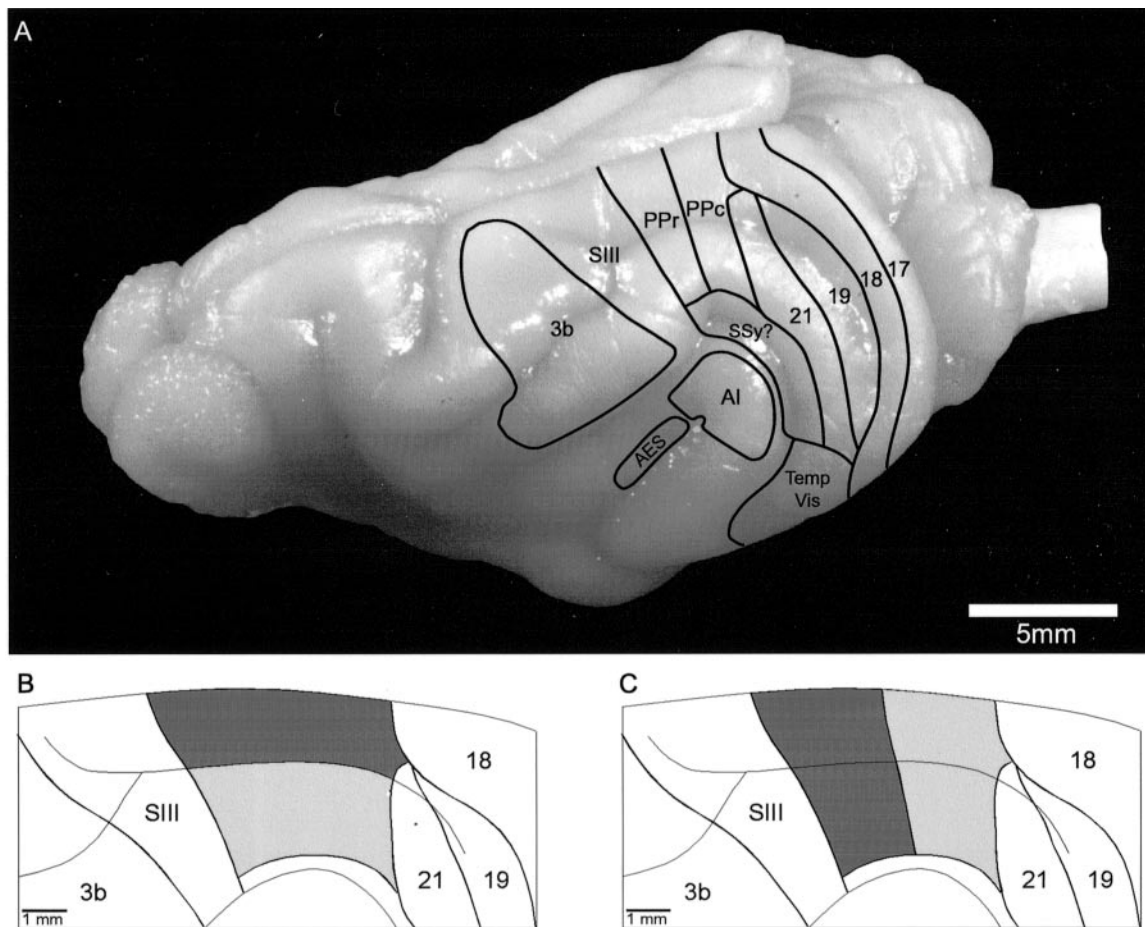


Figure 14. Summary figure of our present understanding of the subdivisions of the visual cortex of the ferret. (A) Dorsolateral photograph of the cortex of the ferret brain with the known area outlined. The locations of suprasylvian, temporal and anterior ectosylvian visual areas are based on preliminary data. (B) Parcellation of posterior parietal cortex based on cytoarchitecture, tangential fiber density in layers 3 and 6, and density of callosal connectivity. (C) Parcellation of the same based on electrophysiological maps, radial fascicle morphology, tangential fiber density in layer 4 and thalamocortical connectivity. We have adopted the scheme shown in (C) in the present study. 3b, primary somatosensory cortex; A1, primary auditory cortex; AES, anterior ectosylvian visual area; PPC, posterior parietal caudal area; PPr, posterior parietal rostral area; SSy?, unexplored suprasylvian visual regions; Temp Vis, temporal visual areas.

speculation seems to be that the organization of posterior parietal cortex in the ferret (and likely the flying fox) unveils a potential organizational plan, from which a complex array of areas, such as are seen in the macaque monkey, might have evolved.

In primates, a cluster of areas anterior, but adjacent, to the medial most part of V2 have been described: the dorsomedial visual areas (Rosa and Tweedale, 2001). In the ferret we did not identify any region which might correspond to the dorsomedial visual areas of primates, although we did not examine the medial wall in detail. Given the present findings and those for the flying fox (Rosa, 1999), it appears likely that the dorsomedial complex of areas is found only in primates.

The Relation of Callosal Connections with Topographic Maps in Parietal Cortex

In a previous study (Manger *et al.*, 2002), we demonstrated the existence of a correlation between callosal connections and retinotopy in areas 17, 18, 19 and 21 of the ferret. In these areas, callosal connections conform to the so-called 'midline rule' discovered in early studies (Berlucchi *et al.*, 1967; Hubel and Wiesel, 1967) and repeatedly confirmed (Innocenti, 1986). The midline rule describes the observation that callosal connections

are preferentially distributed in regions representing the central portions of the visual field. However, due to the complex organization of posterior parietal cortex in the macaque and the lack of reports of topographic maps, these relationships have not been explicitly studied in these higher-order areas.

Our findings indicate that the 'midline rule' is violated in the posterior parietal cortex of the ferret, since callosal connections were distributed throughout PPC and PPr with no relationship to the representations of either visual or somatosensory midlines. Thus, it appears that neurons representing the periphery of the visual field, or the body, can interact through the corpus callosum. This predicts the existence of large receptive fields stretching across the vertical meridian and of split receptive fields restricted to the temporal portions of the hemifields, as described in a fraction of parietal cortex neurons of the monkey (Motter and Mountcastle, 1981). Such receptive fields were not noted in the present study. This might be due to the nature of our study, based on multi-unit activity in anesthetized animals versus single-unit, awake-behaving primates. However, the receptive fields mapped here are similar to the Class II visual neurons of monkeys, which have smaller, usually contralateral receptive fields, with a sensitive locus in the middle of the receptive field (Mountcastle, 1995). Moreover, it has been noted that ~80% of

somatic neurons in the posterior parietal cortex of the monkey are responsive to passive stimulation (Lacquaniti *et al.*, 1995); thus, in our preparation we may have been recording from clusters of passively responsive Class II type visual and/or somatic neurons.

Our findings suggest that in posterior parietal areas a different relationship might exist between callosal connections and sensory maps. Cortical regions representing the upper visual field (in both PPc and PPr of the ferret) as well as whisker pad and jaws (in PPr) appear to have somewhat denser callosal connections, originating from larger neurons, than those cortical regions representing the lower visual field (in both PPc and PPr), or postcranial body (in PPr). It would be important to know if these anatomical differences correlate with electrophysiological properties such as strength of target activation and conduction velocity.

The Relation between Visual and Somatosensory Maps in Posterior Parietal Cortex

In the posterior parietal cortex of the ferret we found two mirror image representations of the visual field. The topography of the caudal field, PPc, is a rough mirror-image of the most medial portions of areas 18 and 21. Thus, the general tenet of reversals of visual representations across cortical areas is maintained. The retinotopy is coarser, but less complex, than that found in occipital areas and there is a bias towards representation of the lower visual field. In contrast, in the occipital extrastriate areas of the ferret, the central visual field is over-represented and the peripheral visual field restricted to islands embedded within the central visual field representation (Manger *et al.*, 2002). The somatic map in area PPr is also coarser than in purely somatosensory areas. The larger receptive fields do not allow for definition of a fine-grade topographic map as in area 3b of the ferret (McLaughlin *et al.*, 1998; Hunt *et al.*, 2000); however, similar to area 3b, the post-cranial body is represented medially and the head laterally. The distal body is represented anteriorly and the proximal body posteriorly. The somatotopic map of PPr is a mirror reversal of that seen in area SIII (Hunt *et al.*, 2000), which, in turn, is a mirror image of the somatotopic map of area 3b.

The relation between the somatosensory and visual maps in PPr is of interest because parietal cortex is believed to be instrumental in establishing congruence between retina-centered and body-centered spatial coordinates, an essential condition for visually guided motor actions (Caminiti *et al.*, 1996; Andersen *et al.*, 1997; Colby and Goldberg, 1999). We found that the representation of the head is coextensive with that of the upper visual field, while those of the forelimb and forepaw are coextensive with the representation of the lower hemifield. A correspondence between visual and somatosensory receptive fields was initially described for the superior colliculus of the mouse (Dräger and Hubel, 1975). The relation between somatic and visual maps we found in PPr is, in general terms, compatible with that between somatic and visual maps in the deep layers of the superior colliculus of the cat and other species (Meredith and Stein, 1996; Meredith *et al.*, 1991). A correspondence between the somatosensory representation of the head and that of the upper visual field was also found in the parietal cortex (area VIP) of the macaque (Duhamel *et al.*, 1998). Here, as in the superior colliculus of the cat (Meredith and Stein, 1996), there appears to be correspondence between location of the receptive field on the mouth and nose and central visual field; more lateral locations on the head correspond to temporal locations in the visual field. However, in the ferret, we found that the more

posterior part of the head was coextensive with the representation of the vertical meridian, at the PPc/PPr border, and that the top of the head corresponded to the uppermost visual field. Within the extensive upper jaw and whisker-pad representation, the more proximal somatosensory regions are closer to the vertical meridian representation and the tip of the muzzle is coextensive with the periphery of the upper visual field. Thus, the visual somatosensory correspondence is altered in comparison to that seen in the colliculus of the cat and area VIP of the macaque monkey.

This difference might be explained in several ways. First, the general tenet of mirror reversals of cortical topographic maps (Kaas, 1997) holds for both the visual and somatic maps in the ferret. Thus, the somatic-visual correspondence we describe might merely be the passive result of topographic map formation, leading to a somatosensory-visual superimposition devoid of any specific functional imperative. Indeed, the large size of the receptive fields and, thus, coarse topography, might obviate the need for any particular congruency between inter-modal maps in the posterior parietal cortex. In the macaque monkey somatic sensory cortex, there is both an area 1 and 2, whereas in the same location in ferret cortex there is a single area, SIII. The additional map in the macaque monkey and, hence, an additional mirror reversal, might explain the similarity in visual-somatic congruency of collicular and cortical multi-modal maps in the monkey that were not found here for the ferret.

Alternatively, and possibly more likely, the collicular and the VIP maps may be thought of as oculo-centered maps, their main functions being accurately to guide saccadic eye movements to the location of the body surface experiencing a tactile stimulus. The posterior parietal cortical map of the ferret might be viewed as a map constructed with body-centered coordinates. In this body-centered scenario, it may be important to maintain congruency between a body part and the portion of the visual space where this body part is most commonly directed when a head/body movement, or attentional shift, is executed, thus requiring a transformation from retinal to body-centered coordinates. Hence the correspondence, in the ferret, of lower visual field with paws, of upper visual field with head and jaws, and of nose, muzzle, and digits with the peripheral visual field.

Notes

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